B.TECH 1ST YEAR | 1ST SEMESTER

Engineering Mathematics

Unit I — Matrices

Complete Study Module · Exam-Oriented · Zero-Skip Step-by-Step Solutions

Subject Code: Engineering Mathematics (MA101)

Coverage: Basics → Rank → Inverse → Systems → Iterative Methods

Solved Examples: 45+ Ultra-Detailed Problems

Exercise Problems: 50 Questions (2-Mark & 5-Mark)

Designed for: Zero-level beginners → Exam toppers

📚 Table of Contents

SECTION 1

Introduction to Matrices

What is a Matrix?

A **matrix** is a rectangular arrangement of numbers, symbols, or expressions arranged in **rows** and **columns**. Think of it like a table with rows going left-to-right and columns going top-to-bottom. Each individual entry in the table is called an **element**.

📖 **FORMAL DEFINITION**

A matrix A of order **m × n** (read: "m by n") is a rectangular array of numbers having **m rows** and **n columns**. It is written as:

```
         [a11  a12  a13  ...  a1n]
         [a21  a22  a23  ...  a2n]
     A = [a31  a32  a33  ...  a3n]
         [ .    .    .         .  ]
         [am1  am2  am3  ...  amn]
```

Here, the element a_{ij} means the element in the **i-th row** and **j-th column**.

Understanding Row and Column Notation

The subscript notation a_{ij} always follows this rule:

- **i** = row number (counted from top)
- **j** = column number (counted from left)

Example: In element a_{23}, i = 2 means second row, j = 3 means third column. So a_{23} is located at row 2, column 3.

Why Do We Study Matrices?

Matrices are the foundation of modern science and engineering. They are used in:

- Solving systems of simultaneous equations (circuits, structures)
- Computer graphics (transformations, rotations)
- Data science and machine learning (neural networks)

- Structural analysis (load distributions)
- Electrical networks (Kirchhoff's laws)

Order of a Matrix

DEFINITION

If a matrix has **m** rows and **n** columns, its order (or size) is written as **m × n**.

- The order always reads: **rows × columns**
- Total number of elements = m × n

Matrix	Rows	Columns	Order	Total Elements
A	2	3	2 × 3	6
B	3	3	3 × 3	9
C	1	4	1 × 4	4
D	4	1	4 × 1	4

SECTION 2

Types of Matrices

Type	Definition	Example
Row Matrix	Only 1 row, n columns. Order = 1 × n.	[1 2 3]
Column Matrix	Only 1 column, m rows. Order = m × 1.	[1] [2] [3]
Square Matrix	m = n (rows = columns).	[1 2] [3 4]
Zero Matrix (Null)	All elements = 0.	[0 0] [0 0]
Identity Matrix (I)	Square matrix, diagonal = 1, rest = 0.	[1 0 0] [0 1 0] [0 0 1]
Diagonal Matrix	Square; non-diagonal elements = 0.	[3 0 0] [0 5 0] [0 0 7]
Scalar Matrix	Diagonal matrix with all diagonal elements equal.	[4 0] [0 4]
Upper Triangular	All elements below diagonal = 0.	[1 2 3] [0 4 5]

Type	Definition	Example
		[0 0 6]
Lower Triangular	All elements above diagonal = 0.	[1 0 0] [2 3 0] [4 5 6]
Symmetric Matrix	$A = A^T$, i.e., $a_{ij} = a_{ji}$.	[1 2 3] [2 4 5] [3 5 6]
Skew-Symmetric	$A = -A^T$, i.e., $a_{ij} = -a_{ji}$; diagonal = 0.	[0 2 -3] [-2 0 1] [3 -1 0]
Singular Matrix	det(A) = 0. Inverse does not exist.	[1 2] [2 4]
Non-Singular Matrix	det(A) ≠ 0. Inverse exists.	[1 2] [3 4]
Orthogonal Matrix	$A \cdot A^T = I$, so $A^{-1} = A^T$.	cos θ, sin θ rotation matrices
Idempotent Matrix	$A^2 = A$.	Projection matrices

💡 **MEMORY TIP**

Identity vs Diagonal vs Scalar: Think of them as Russian dolls — Identity is inside Scalar, Scalar is inside Diagonal, Diagonal is inside Square.

Identity ⊂ Scalar ⊂ Diagonal ⊂ Square

SECTION 3

Matrix Operations

3.1 Matrix Addition and Subtraction

RULE

Two matrices can be added or subtracted **only if they have the same order**. You simply add (or subtract) the corresponding elements.

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are both $m \times n$ matrices, then:

$\mathbf{A + B = [a_{ij} + b_{ij}]}$ and $\mathbf{A - B = [a_{ij} - b_{ij}]}$

3.2 Scalar Multiplication

When a scalar (a plain number) k multiplies a matrix A, every element of A is multiplied by k.

$$k \cdot A = [k \cdot a_{ij}]$$

3.3 Matrix Multiplication

CONDITION FOR MULTIPLICATION

Matrix A (of order $m \times n$) can multiply matrix B (of order $p \times q$) **only when $n = p$** (columns of A = rows of B). The resulting matrix $C = A \times B$ has order $\mathbf{m \times q}$.

HOW TO COMPUTE EACH ELEMENT

Each element c_{ij} of the product matrix C is found by taking the **dot product** of the i-th row of A with the j-th column of B:

$$c_{ij} = a_{i1} \cdot b_{1j} + a_{i2} \cdot b_{2j} + a_{i3} \cdot b_{3j} + \ldots + a_{in} \cdot b_{nj}$$

⚠ IMPORTANT WARNING

Matrix multiplication is **NOT commutative**. In general, A × B ≠ B × A. Always check the order before multiplying.

3.4 Transpose of a Matrix

DEFINITION

The **transpose** of a matrix A, written as A^T (or A'), is obtained by **interchanging its rows and columns**. The element at position (i, j) in A moves to position (j, i) in A^T.

If A is m × n, then A^T is n × m.

Properties of Transpose

Property	Statement
$(A^T)^T = A$	Double transpose gives back original
$(A + B)^T = A^T + B^T$	Transpose distributes over addition
$(kA)^T = k \cdot A^T$	Scalar can be factored out
$(AB)^T = B^T A^T$	Order reverses during transpose of product

SECTION 4

Rank of a Matrix – Echelon Form Method

What is the Rank of a Matrix?

📖 **DEFINITION**

The **rank** of a matrix A, written as ρ(A) or rank(A), is defined as the **maximum number of linearly independent rows** (or columns) in the matrix. Equivalently, it is the **order of the largest non-zero minor** of the matrix.

💡 **SIMPLE INTUITION**

Think of rank as "how many rows actually carry new, useful information." If one row is a multiple of another, it adds no new information — so it doesn't count toward the rank.

Row Echelon Form (REF) – What does it look like?

DEFINITION

A matrix is in **Row Echelon Form** when:

1. All zero rows (if any) are at the **bottom**.
2. The first non-zero element of each row (called the **pivot** or **leading entry**) is to the **right** of the pivot in the row above it.
3. All elements **below each pivot** are zero.

Elementary Row Operations (ERO)

To convert any matrix to echelon form, we use only three types of operations. These operations do NOT change the rank of the matrix.

Operation	Symbol	Meaning
Row interchange	$R_i \leftrightarrow R_j$	Swap row i with row j
Row scaling	$R_i \to k \cdot R_i$	Multiply all elements of row i by k ($k \neq 0$)

Operation	Symbol	Meaning
Row replacement	$R_i \to R_i + k \cdot R_j$	Add k times row j to row i

FINDING RANK FROM ECHELON FORM

Rank = Number of non-zero rows in the Row Echelon Form

After converting the matrix to REF, simply count how many rows are NOT entirely zero. That count is the rank.

Step-by-Step Strategy to Find Rank

1. **Identify the pivot column** (leftmost non-zero column).
2. **Bring a non-zero element to the top** of that column (using row swap if needed).
3. **Eliminate all elements below the pivot** using row replacement operations.
4. **Move to the next sub-matrix** (ignore the pivot row and column).
5. **Repeat** until the matrix is in echelon form.
6. **Count non-zero rows** → that is the rank.

SECTION 5

Rank of a Matrix – Normal Form Method

📖 **NORMAL FORM (CANONICAL FORM)**

Any matrix A of rank r can be reduced to its **Normal Form** (also called Canonical Form or Hermite Normal Form) using both **elementary row operations AND elementary column operations**. The normal form looks like:

```
Normal Form = [ I_r   | 0 ]
              [  0    | 0 ]
```

Where I_r is an r × r identity matrix in the top-left corner, and all other positions are filled with zeros.

Elementary Column Operations

Just like row operations, we can perform the same three types of operations on columns:

Operation	Symbol	Meaning
Column interchange	$C_i \leftrightarrow C_j$	Swap column i with column j
Column scaling	$C_i \rightarrow k \cdot C_i$	Multiply all elements of column i by k (k ≠ 0)
Column replacement	$C_i \rightarrow C_i + k \cdot C_j$	Add k times column j to column i

FINDING RANK FROM NORMAL FORM

Rank = r = Size of the identity matrix block I_r in the normal form

If the normal form has a 3 × 3 identity block, then rank = 3.

📝 **NOTE**

The Normal Form method is more powerful than echelon form because using column operations often makes the computation easier. However, it is more work. Both methods give the same rank.

SECTION 6

Cauchy–Binet Formula

📜 **STATEMENT OF CAUCHY–BINET FORMULA**

Let A be an m × n matrix and B be an n × p matrix (both are **not necessarily square**). If m = p, then the determinant of the product matrix AB (which is m × m) can be computed as:

```
det(AB) = Sum over all r-subsets S of {1,2,...,n} of:
           [A(S)] × [B(S)]

where r = m = p,
A(S) = minor of A formed by columns in set S,
B(S) = minor of B formed by rows in set S.
```

Special Case: Square Matrices

```
If A and B are both n × n square matrices:

det(AB) = det(A) × det(B)
```

This special case is the most commonly used result. It says: *the determinant of a product equals the product of determinants.*

Important Consequences

Result	Explanation
det(AB) = det(A)·det(B)	For square matrices of same order
$\det(A^n) = [\det(A)]^n$	Determinant of power equals power of determinant
If det(A) = 0 then det(AB) = 0	Singular matrix makes the product singular
$\det(A \cdot A^{-1}) = \det(I) = 1$	So $\det(A) \cdot \det(A^{-1}) = 1$

📝 **EXAM NOTE**

In B.Tech examinations, the Cauchy-Binet formula is usually asked as a **statement question**. You need to state the formula clearly and know its special case det(AB) = det(A)·det(B) for square matrices. Full proof is generally not required.

SECTION 7

Inverse of a Matrix – Adjoint Method

What is the Inverse of a Matrix?

📖 **DEFINITION**

The **inverse** of a square matrix A (written as A^{-1}) is the matrix such that:

$$A \cdot A^{-1} = A^{-1} \cdot A = I \text{ (Identity Matrix)}$$

The inverse exists **only when det(A) ≠ 0** (i.e., the matrix is non-singular).

Steps to Find Inverse Using Adjoint Method

1. **Compute det(A)** — if it is 0, inverse does not exist. Stop.
2. **Find the Cofactor Matrix** of A.
3. **Find the Adjoint** = Transpose of Cofactor Matrix.
4. **Apply the formula:** $A^{-1} = (1/\det(A)) \times \text{adj}(A)$

What is a Cofactor?

DEFINITION

The **cofactor C_{ij}** of element a_{ij} is:

$$C_{ij} = (-1)^{i+j} \times M_{ij}$$

Where M_{ij} is the **minor** of a_{ij} = determinant of the submatrix obtained by deleting row i and column j.

The sign pattern **$(-1)^{i+j}$** follows a checkerboard pattern:

```
[+  −  +]
[−  +  −]
```

```
[+   −   +]
```

The Adjoint (Adjugate)

$$\text{adj}(A) = [C_{ij}]^T = \text{Transpose of the Cofactor Matrix}$$

⚠ **COMMON MISTAKE**

Students often forget to **transpose** the cofactor matrix to get the adjoint. Remember: Cofactor matrix ≠ Adjoint. Adjoint = **Transpose** of cofactor matrix.

SECTION 8

Inverse of a Matrix – Gauss-Jordan Method

The Big Idea

The Gauss-Jordan method finds the inverse by applying row operations simultaneously to the matrix A and the identity matrix I. The key insight is:

```
Start with [A | I] → Apply Row Operations → Reach [I | A⁻¹]
```

We keep transforming the left side (A) into the identity matrix using row operations, and whatever the same operations do to the right side (I), that becomes A^{-1}.

Detailed Steps

1. **Write the augmented matrix** $[A \mid I_n]$ where I_n is the $n \times n$ identity matrix.
2. **Use elementary row operations** to convert the left part A into the identity matrix I.
3. **Apply every row operation to both sides** simultaneously.
4. When the left part becomes I, the **right part automatically becomes A^{-1}**.
5. If at any point a row of zeros appears on the left side, **the matrix is singular** (inverse doesn't exist).

💡 **WHY THIS WORKS**

Each elementary row operation is equivalent to multiplying by an elementary matrix E on the left. If we perform operations $E_1, E_2, \ldots, E_k$ to convert A to I, then:

$E_k \ldots E_2 E_1 \cdot A = I \longrightarrow$ So $E_k \ldots E_2 E_1 = A^{-1}$

And $E_k \ldots E_2 E_1 \cdot I = A^{-1} \cdot I = A^{-1}$. That's exactly what appears on the right side!

SECTION 9

System of Linear Equations – Introduction

What is a System of Linear Equations?

A system of linear equations is a collection of two or more equations, each involving the same set of variables, where every variable appears only to the first power (no x^2, xy terms, etc.).

GENERAL FORM

```
a11·x1 + a12·x2 + ... + a1n·xn = b1
a21·x1 + a22·x2 + ... + a2n·xn = b2
.        .                .      .
am1·x1 + am2·x2 + ... + amn·xn = bm
```

In matrix form: **AX = B**, where A is the coefficient matrix, X is the column of unknowns, and B is the column of constants.

Classification of Systems

Type	Condition	Solutions
Consistent (Unique Solution)	rank(A) = rank([A|B]) = n	Exactly one solution
Consistent (Infinite Solutions)	rank(A) = rank([A|B]) = r < n	Infinite solutions (n−r free variables)
Inconsistent (No Solution)	rank(A) ≠ rank([A|B])	No solution exists

📜 ROUCHÉ–CAPELLI THEOREM (CONSISTENCY THEOREM)

The system AX = B is **consistent** (has at least one solution) **if and only if** rank(A) = rank([A|B]), where [A|B] is the augmented matrix.

- If rank(A) = rank([A|B]) = n → **Unique solution**
- If rank(A) = rank([A|B]) < n → **Infinite solutions** (n − rank unknowns are free)

- If rank(A) ≠ rank([A | B]) → **Inconsistent** (no solution)

SECTION 10

Homogeneous Systems

What is a Homogeneous System?

📖 **DEFINITION**

A system AX = B is called **homogeneous** when B = 0 (the right-hand side is all zeros). The system becomes AX = 0.

KEY FACT

A homogeneous system **always has at least one solution**: the **trivial solution** X = 0 (all unknowns = 0). This is why homogeneous systems are always consistent.

When Does AX = 0 Have Non-Trivial Solutions?

Condition	Type of Solution
rank(A) = n (number of unknowns)	Only trivial solution X = 0
rank(A) = r < n	Infinite non-trivial solutions; (n − r) free variables
For square n × n matrix: det(A) ≠ 0	Only trivial solution
For square n × n matrix: det(A) = 0	Non-trivial solutions exist

SECTION 11

Non-Homogeneous Systems

📖 **DEFINITION**

When $B \neq 0$ in $AX = B$, the system is called **non-homogeneous**. These systems may have no solution, a unique solution, or infinitely many solutions, depending on the ranks.

Complete Analysis Framework

Given $AX = B$ with m equations and n unknowns:

1. Form the augmented matrix $[A \mid B]$.
2. Reduce to row echelon form using row operations.
3. Find rank(A) and rank([A|B]).
4. Apply the Rouché-Capelli Theorem to classify the solution.
5. If consistent, find the solution(s).

📝 **HOW TO EXPRESS INFINITE SOLUTIONS**

When $\text{rank}(A) = r < n$, there are $(n - r)$ **free variables**. You can assign any value to the free variables and express the other variables in terms of them. This gives the general solution.

Example: If $n = 3$ and $r = 2$, then one variable is free. Say $x_3 = t$. Then express x_1 and x_2 in terms of t.

SECTION 12

Gauss Elimination Method

What is Gauss Elimination?

Gauss Elimination (also called Gaussian Elimination) is a systematic method to solve a system of linear equations AX = B. It converts the augmented matrix [A|B] into an upper triangular form using row operations, then solves by **back substitution**.

Phase 1: Forward Elimination

1. Write the augmented matrix [A | B].
2. Find the first pivot (top-left non-zero element).
3. Use row operations to make all elements **below the pivot = 0**.
4. Move to the next pivot (next row, next column) and repeat.
5. Continue until the matrix is upper triangular.

Phase 2: Back Substitution

1. Start from the **last equation** (which has only one unknown).
2. Solve for that unknown directly.
3. Substitute its value into the second-to-last equation.
4. Solve for the next unknown.
5. Continue moving upward until all unknowns are found.

💡 GAUSS VS GAUSS-JORDAN

Gauss Elimination: Reduces to *upper triangular* form, then uses back substitution.

Gauss-Jordan Elimination: Reduces to *reduced row echelon form* (diagonal 1s), reads off solutions directly — no back substitution needed.

SECTION 13

Iterative Methods – Jacobi & Gauss-Seidel

Why Iterative Methods?

For large systems with many equations, direct methods (like Gauss Elimination) become computationally expensive. Iterative methods start with an initial guess and **repeatedly improve the solution** until it converges to the true answer.

13.1 Jacobi Method

IDEA

In the Jacobi method, each unknown is solved from its equation, **using only values from the previous iteration**. All unknowns are updated simultaneously at the end of each iteration.

SETUP – REARRANGING EQUATIONS

Given the system:

```
a11·x1 + a12·x2 + a13·x3 = b1   →   x1 = (b1 − a12·x2 − a13·x3) / a11
a21·x1 + a22·x2 + a23·x3 = b2   →   x2 = (b2 − a21·x1 − a23·x3) / a22
a31·x1 + a32·x2 + a33·x3 = b3   →   x3 = (b3 − a31·x1 − a32·x2) / a33
```

Starting with initial guess x1(0) = x2(0) = x3(0) = 0, substitute into the right side to get the next iteration values.

⚠ CONVERGENCE CONDITION

Jacobi (and Gauss-Seidel) methods **converge** if the system is **diagonally dominant**, meaning for each equation i:

$$|a_{ii}| > \text{sum of } |a_{ij}| \text{ for all } j \neq i$$

i.e., the diagonal element must be larger in absolute value than the sum of all other elements in that row.

13.2 Gauss-Seidel Method

> **IDEA**
>
> Gauss-Seidel is an improvement of Jacobi. Each unknown, as soon as it is computed in the **current iteration**, is **immediately used** in the same iteration (not waiting for the next round). This makes it converge faster.

KEY DIFFERENCE: JACOBI VS GAUSS-SEIDEL

Feature	Jacobi Method	Gauss-Seidel Method
Update rule	Use all old values	Use newest available values
Speed of convergence	Slower	Faster (roughly 2×)
Storage needed	Two sets of values	One set (overwrite)
Convergence condition	Diagonal dominance	Diagonal dominance

SECTION 14

Solved Examples – Ultra Step-by-Step Solutions

Example 1 Find the rank of a 2×3 matrix using Echelon Form Easy

📋 **GIVEN**

```
    [ 1  2  3 ]
A = [ 2  4  6 ]
```

Find rank(A).

💡 **CONCEPT USED**

We use the Row Echelon Form method. We apply elementary row operations to convert A into a form where we can count non-zero rows. The rank equals the number of non-zero rows in echelon form.

Step 1 – Understand the Matrix

We have a 2 × 3 matrix. Maximum possible rank = min(2, 3) = 2. But we must check if the rows are linearly independent.

Look at Row 2: [2 4 6]. Is this related to Row 1: [1 2 3]? Yes — Row 2 = 2 × Row 1! So Row 2 carries no new information.

Why does this matter? Rows that are multiples of each other don't add to the rank. We expect rank = 1.

Step 2 – Write the Matrix and Identify the Pivot

```
A = [ 1  2  3 ]   ← Pivot is a11 = 1 (first non-zero in row 1)
    [ 2  4  6 ]
```

The pivot is the element 1 in row 1, column 1. Our goal is to make all elements **below this pivot equal to zero**.

Why make elements below pivot zero? Because that's the definition of echelon form — each column can only have one leading non-zero element (the pivot), and everything below must be zero.

Step 3 – Plan the Operation (Before Doing It)

We need to eliminate the element **2** in Row 2, Column 1.

To make it zero, we will subtract 2 times Row 1 from Row 2.

Operation: $\mathbf{R_2 \rightarrow R_2 - 2 \cdot R_1}$

Why subtract 2 times R1? Because element a21 = 2 and pivot a11 = 1. To make a21 zero: 2 − 2×1 = 0. We subtract (a21/a11) × R1 from R2.

Step 4 – Perform the Operation

Operation: R2 → R2 − 2·R1

```
New R2, element 1: 2 − 2×(1) = 2 − 2 = 0
New R2, element 2: 4 − 2×(2) = 4 − 4 = 0
New R2, element 3: 6 − 2×(3) = 6 − 6 = 0
```

```
Result: A ~ [ 1   2   3 ]
            [ 0   0   0 ]
```

The entire second row became zero. This confirms that Row 2 was completely dependent on Row 1.

Step 5 – Read the Rank

The matrix is now in echelon form. Count the non-zero rows:

- Row 1: [1 2 3] → NOT zero ✓
- Row 2: [0 0 0] → All zeros ✗

Number of non-zero rows = **1**

✅ FINAL ANSWER

rank(A) = 1

📘 **WHAT YOU LEARNED**

When one row of a matrix is an exact multiple of another row, the rank decreases by 1. Always check for row dependency before assuming maximum rank. Echelon form makes this dependency visible as a row of zeros.

Example 2 Find the rank of a 3×3 matrix **Easy**

📋 **GIVEN**

```
    [ 1   2  -1 ]
A = [ 2   4   3 ]
    [ 3   6   2 ]
```

💡 **CONCEPT USED**

Row Echelon Form. We apply operations to eliminate elements below each pivot column by column.

Step 1 – Identify the First Pivot

The first pivot is a_{11} = 1 (row 1, col 1). We need to eliminate a_{21} = 2 and a_{31} = 3.

The pivot is the first non-zero leading element. We use it to zero out everything below it in the same column.

Step 2 – Eliminate Column 1 Below Pivot

We need two operations:

Operation 1: R2 → R2 − 2·R1 (to make a21 = 0)

```
New R2 elem 1: 2 − 2(1) = 0
New R2 elem 2: 4 − 2(2) = 0
New R2 elem 3: 3 − 2(−1) = 3 + 2 = 5
```

Operation 2: R3 → R3 − 3·R1 (to make a31 = 0)

```
New R3 elem 1: 3 − 3(1) = 0
New R3 elem 2: 6 − 3(2) = 0
New R3 elem 3: 2 − 3(−1) = 2 + 3 = 5
```

```
After Step 2:
A ~ [ 1  2  -1 ]
    [ 0  0   5 ]
    [ 0  0   5 ]
```

Step 3 – Move to Next Pivot Position

The next pivot should be at row 2, column 2. But a_{22} = 0! Column 2 has no non-zero entry below the pivot row. We skip to column 3.

The new pivot is a_{23} = 5. We need to eliminate a_{33} = 5 below it.

When an entire sub-column is zero, we skip to the next column. The pivot moves right but stays in the same row.

Step 4 – Eliminate Column 3 Below New Pivot

Operation: R3 → R3 − R2 (to make a33 = 0 below pivot at a23)

```
New R3 elem 1: 0 − 0 = 0
New R3 elem 2: 0 − 0 = 0
New R3 elem 3: 5 − 5 = 0
```

```
Final Echelon Form:
A ~ [ 1   2   -1 ]
    [ 0   0    5 ]
    [ 0   0    0 ]
```

Step 5 – Count Non-Zero Rows

- Row 1: [1 2 -1] → non-zero ✓
- Row 2: [0 0 5] → non-zero ✓
- Row 3: [0 0 0] → all zero ✗

Non-zero rows = 2

✅ FINAL ANSWER

rank(A) = 2

📘 **WHAT YOU LEARNED**

When a pivot column has a zero in the current row, skip to the next column. The echelon form may have "missing" pivots in some columns. The rank is still the count of non-zero rows, not the number of columns with pivots.

Example 3 Find the rank of a 4×4 matrix (Medium) Medium

📋 **GIVEN**

```
    [ 1  2  3  4 ]
A = [ 2  3  4  5 ]
    [ 3  4  5  6 ]
    [ 4  5  6  7 ]
```

💡 **CONCEPT USED**

This matrix has a pattern where each row = previous row + [1,1,1,1]. Let's apply echelon form and discover the actual rank.

Step 1 – Eliminate Column 1

R2 → R2 − 2·R1

`[2−2, 3−4, 4−6, 5−8] = [0, −1, −2, −3]`

R3 → R3 − 3·R1

`[3−3, 4−6, 5−9, 6−12] = [0, −2, −4, −6]`

R4 → R4 − 4·R1

`[4−4, 5−8, 6−12, 7−16] = [0, −3, −6, −9]`

```
After Step 1:
A ~ [ 1   2   3   4 ]
    [ 0  −1  −2  −3 ]
    [ 0  −2  −4  −6 ]
    [ 0  −3  −6  −9 ]
```

Step 2 – Eliminate Column 2 (Pivot = −1 at row 2, col 2)

R3 → R3 − 2·R2 (factor = −2/−1 = 2)

`[0−0, −2−(−2), −4−(−4), −6−(−6)] = [0, 0, 0, 0]`

R4 → R4 − 3·R2 (factor = −3/−1 = 3)

`[0−0, −3−(−3), −6−(−6), −9−(−9)] = [0, 0, 0, 0]`

```
After Step 2:
A ~ [ 1   2   3   4 ]
    [ 0  −1  −2  −3 ]
    [ 0   0   0   0 ]
    [ 0   0   0   0 ]
```

Step 3 – Count Non-Zero Rows

- Row 1: [1 2 3 4] → non-zero ✓
- Row 2: [0 −1 −2 −3] → non-zero ✓
- Row 3: all zeros ✗
- Row 4: all zeros ✗

✅ FINAL ANSWER

rank(A) = 2

📘 **WHAT YOU LEARNED**

Even a 4×4 matrix can have rank 2 if the rows form an arithmetic progression (each row differs from the previous by a constant). Pattern recognition saves work. Always reduce fully before counting.

Example 4 **Find the inverse of a 3×3 matrix using Gauss-Jordan Method** Medium

GIVEN

```
    [ 2  1  0 ]
A = [ 1  2  1 ]
    [ 0  1  2 ]
```

Find A^{-1} using Gauss-Jordan Method.

CONCEPT USED

Gauss-Jordan: We write [A | I] and transform the left A into I using row operations. The right side automatically becomes A^{-1}.

Step 1 – Write the Augmented Matrix [A | I]

```
[ 2  1  0 | 1  0  0 ]
[ 1  2  1 | 0  1  0 ]
[ 0  1  2 | 0  0  1 ]
```

We place the 3×3 identity matrix on the right. Every row operation we do to the left side is applied identically to the right side. When left becomes I, right becomes A^{-1}.

Step 2 – Make Pivot at (1,1) = 1 by Swapping R1 and R2

Currently a_{11} = 2. It's easier to work with pivot = 1. Since a_{21} = 1, let's swap R1 and R2.

R1 ↔ R2

```
[ 1  2  1 | 0  1  0 ]
[ 2  1  0 | 1  0  0 ]
[ 0  1  2 | 0  0  1 ]
```

Having pivot = 1 makes subsequent arithmetic simpler and avoids fractions for as long as possible.

Step 3 – Eliminate Column 1 Below Pivot

R2 → R2 − 2·R1

```
Left: [2-2(1), 1-2(2), 0-2(1)] = [0, -3, -2]
Right: [1-2(0), 0-2(1), 0-2(0)] = [1, -2, 0]
```

```
[ 1   2   1 | 0   1   0 ]
[ 0  -3  -2 | 1  -2   0 ]
[ 0   1   2 | 0   0   1 ]
```

Row 3 starts with 0 in column 1, so we don't need to touch it for column 1 elimination.

Step 4 – Make Pivot at (2,2) Easier: Swap R2 and R3

Currently pivot position (2,2) has −3. Row 3 has +1 at (3,2). Swapping gives pivot = 1 again.

R2 ↔ R3

```
[ 1   2   1 | 0   1   0 ]
[ 0   1   2 | 0   0   1 ]
[ 0  -3  -2 | 1  -2   0 ]
```

Step 5 – Eliminate Column 2 (Above and Below Pivot)

R1 → R1 − 2·R2 (eliminate above pivot in col 2)

```
Left: [1-0, 2-2, 1-4] = [1, 0, -3]
Right: [0-0, 1-0, 0-2] = [0, 1, -2]
```

R3 → R3 + 3·R2 (eliminate below pivot in col 2)

```
Left: [0+0, -3+3, -2+6] = [0, 0, 4]
Right: [1+0, -2+0, 0+3] = [1, -2, 3]
```

```
[ 1  0  -3 | 0   1  -2 ]
[ 0  1   2 | 0   0   1 ]
[ 0  0   4 | 1  -2   3 ]
```

Step 6 – Make Pivot at (3,3) = 1 by Scaling R3

R3 → (1/4)·R3

```
Left: [0, 0, 1]
Right: [1/4, -1/2, 3/4]
```

```
[ 1   0   -3 | 0       1     -2    ]
[ 0   1    2 | 0       0      1    ]
[ 0   0    1 | 1/4   -1/2    3/4   ]
```

Step 7 – Eliminate Column 3 (Above Pivot)

R1 → R1 + 3·R3 (to eliminate −3 in R1, col 3)

```
Left: [1+0, 0+0, -3+3] = [1, 0, 0]
Right: [0+3/4, 1+(-3/2), -2+9/4] = [3/4, -1/2, 1/4]
```

R2 → R2 − 2·R3 (to eliminate 2 in R2, col 3)

```
Left: [0, 1, 0]
Right: [0-1/2, 0+1, 1-3/2] = [-1/2, 1, -1/2]
```

```
[ 1   0   0 |   3/4    -1/2    1/4   ]
[ 0   1   0 | -1/2      1     -1/2   ]
[ 0   0   1 |   1/4    -1/2    3/4   ]
```

Step 8 – Read Off the Inverse

The left side is now the identity matrix I. The right side is A^{-1}.

✅ FINAL ANSWER

```
              [ 3/4    -1/2    1/4 ]
A-1    =      [-1/2     1     -1/2 ]
              [ 1/4    -1/2    3/4 ]
```

📘 **WHAT YOU LEARNED**

Gauss-Jordan inverse requires patience and careful arithmetic. Swapping rows to bring smaller pivot values (like 1) upward reduces fraction errors. Always apply every operation to BOTH sides of the augmented matrix simultaneously.

Example 5 Solve a 3×3 system using Gauss Elimination Medium

📋 **GIVEN SYSTEM**

```
2x + y + z  = 10
3x + 2y + 3z = 18
x + 4y + 9z = 16
```

💡 **CONCEPT USED**

Gauss Elimination: Convert to upper triangular form using forward elimination, then solve from bottom to top using back substitution.

Step 1 – Write the Augmented Matrix [A | B]

```
[ 2  1  1 | 10 ]
[ 3  2  3 | 18 ]
[ 1  4  9 | 16 ]
```

The augmented matrix combines the coefficient matrix A with the constants column B. Every row operation on this matrix corresponds to the same operation on the original equations.

Step 2 – Swap R1 and R3 to Get Pivot = 1

R1 ↔ R3

```
[ 1  4  9 | 16 ]
[ 3  2  3 | 18 ]
[ 2  1  1 | 10 ]
```

Step 3 – Eliminate Column 1 Below Pivot

R2 → R2 − 3·R1

```
[3−3, 2−12, 3−27 | 18−48] = [0, −10, −24 | −30]
```

R3 → R3 − 2·R1

```
[2−2, 1−8, 1−18 | 10−32] = [0, −7, −17 | −22]
```

```
[ 1    4    9   |  16 ]
[ 0  -10  -24   | -30 ]
[ 0   -7  -17   | -22 ]
```

Step 4 – Simplify R2 (Divide by −2 to reduce numbers)

R2 → R2 / (−2)

```
[0, 5, 12 | 15]
```

```
[ 1    4    9   |  16 ]
[ 0    5   12   |  15 ]
[ 0   -7  -17   | -22 ]
```

Step 5 – Eliminate Column 2 Below Pivot (Pivot = 5 at row 2)

R3 → R3 + (7/5)·R2 — to eliminate a32 = −7

```
Factor = -(-7)/5 = 7/5
R3[1]: 0 + (7/5)(0) = 0
R3[2]: -7 + (7/5)(5) = -7 + 7 = 0
R3[3]: -17 + (7/5)(12) = -17 + 84/5 = -85/5 + 84/5 = -1/5
R3[B]: -22 + (7/5)(15) = -22 + 21 = -1
```

```
[ 1    4    9   |  16  ]
[ 0    5   12   |  15  ]
[ 0    0  -1/5  |  -1  ]
```

Step 6 – Simplify R3

R3 → R3 × (−5)

```
[0, 0, 1 | 5]
```

```
Upper Triangular Form:
[ 1    4    9   |  16 ]
[ 0    5   12   |  15 ]
[ 0    0    1   |   5 ]
```

Step 7 – Back Substitution (Bottom to Top)

From Row 3: $z = 5$

From Row 2: $5y + 12z = 15 \rightarrow 5y + 12(5) = 15 \rightarrow 5y + 60 = 15 \rightarrow 5y = -45 \rightarrow y = -9$

From Row 1: $x + 4y + 9z = 16 \rightarrow x + 4(-9) + 9(5) = 16 \rightarrow x - 36 + 45 = 16 \rightarrow x + 9 = 16 \rightarrow x = 7$

✅ FINAL ANSWER

$x = 7, \quad y = -9, \quad z = 5$

📘 **WHAT YOU LEARNED**

Gauss elimination has two phases: forward (create triangular form) and backward (solve bottom-up). Intermediate simplification (like dividing a row by a common factor) reduces arithmetic errors significantly.

Example 6

Test consistency and find solution using Rouché-Capelli Theorem

Medium

GIVEN SYSTEM

```
x + y + z   = 6
x + 2y + 3z = 14
x + 4y + 9z = 36
```

CONCEPT USED

We use the Rouché-Capelli (Consistency) Theorem: Compare rank(A) with rank([A|B]) to determine if the system has a solution, and how many.

Step 1 – Form the Augmented Matrix [A | B]

```
[A | B] = [ 1   1   1 | 6   ]
          [ 1   2   3 | 14 ]
          [ 1   4   9 | 36 ]
```

Step 2 – Apply Row Operations

R2 → R2 − R1

```
[0, 1, 2 | 8]
```

R3 → R3 − R1

```
[0, 3, 8 | 30]
```

```
[ 1   1   1 | 6   ]
[ 0   1   2 | 8   ]
[ 0   3   8 | 30 ]
```

Step 3 – Eliminate Column 2 Below Pivot

R3 → R3 − 3·R2

```
[0, 3-3, 8-6 | 30-24] = [0, 0, 2 | 6]
```

```
[ 1   1   1 | 6 ]
[ 0   1   2 | 8 ]
[ 0   0   2 | 6 ]
```

Step 4 – Determine Ranks

rank(A) = non-zero rows of A = 3 (all three rows non-zero in echelon form of A)

rank([A|B]) = non-zero rows of augmented matrix = 3

Since rank(A) = rank([A|B]) = 3 = n (number of unknowns), the system is **consistent with a unique solution**.

Step 5 – Back Substitution

Row 3: $2z = 6 \rightarrow z = 3$

Row 2: $y + 2(3) = 8 \rightarrow y = 2$

Row 1: $x + 2 + 3 = 6 \rightarrow x = 1$

FINAL ANSWER

Consistent. Unique solution: $x = 1$, $y = 2$, $z = 3$

Example 7 Show that a system is Inconsistent Medium

GIVEN SYSTEM

```
x + y + z  = 3
2x + 2y + 2z = 7
3x + 3y + 3z = 10
```

CONCEPT USED

If rank(A) ≠ rank([A|B]), the system is inconsistent (no solution). This occurs when the constants conflict with the structure of the equations.

Step 1 – Form Augmented Matrix

```
[ 1  1  1 | 3 ]
[ 2  2  2 | 7 ]
[ 3  3  3 | 10 ]
```

Step 2 – Apply Operations

R2 → R2 − 2·R1

```
[0, 0, 0 | 7 − 6] = [0, 0, 0 | 1]
```

R3 → R3 − 3·R1

```
[0, 0, 0 | 10 − 9] = [0, 0, 0 | 1]
```

```
[ 1  1  1 | 3 ]
[ 0  0  0 | 1 ]
[ 0  0  0 | 1 ]
```

Step 3 – Interpret Row 2

Row 2 now reads: **0·x + 0·y + 0·z = 1**, which simplifies to **0 = 1**.

This is a **contradiction** — it can never be true!

Therefore rank([A|B]) = 2 (rows 1 and 2 are non-zero), but rank(A) = 1 (only row 1 is non-zero in A-part).

Since rank(A) ≠ rank([A|B]), by the Rouché-Capelli Theorem, the system is **inconsistent**.

✅ FINAL ANSWER

The system is INCONSISTENT – No solution exists

📘 **WHAT YOU LEARNED**

When multiple equations are proportional in coefficients but NOT in constants (here all LHS are multiples of the first equation, but the RHS are 3, 7, 10 which are not proportional), a contradiction arises. The system is inconsistent.

Example 8 System with Infinite Solutions (Homogeneous-like) Medium

📋 **GIVEN SYSTEM**

```
x + 2y + 3z = 6
2x + 4y + 6z = 12
3x + 6y + 9z = 18
```

Step 1 – Form Augmented Matrix and Reduce

```
[ 1  2  3 | 6  ]
[ 2  4  6 | 12 ]
[ 3  6  9 | 18 ]
```

R2 → R2 − 2·R1 and R3 → R3 − 3·R1

```
Both give [0, 0, 0 | 0]
```

```
[ 1  2  3 | 6 ]
[ 0  0  0 | 0 ]
[ 0  0  0 | 0 ]
```

Step 2 – Determine Ranks and Number of Free Variables

rank(A) = 1, rank([A|B]) = 1. Since they're equal, system is consistent.

Number of unknowns = n = 3. Rank r = 1. Free variables = n − r = 3 − 1 = 2.

Step 3 – Express General Solution

The only equation is: x + 2y + 3z = 6. Let y = s and z = t (free parameters, any real value).

Then: x = 6 − 2s − 3t

General Solution:

```
x = 6 - 2s - 3t
y = s
z = t
where s, t can be any real numbers.
```

FINAL ANSWER

Consistent with Infinite Solutions: $x = 6 - 2s - 3t$, $y = s$, $z = t$

Example 9 Solve using Jacobi Iteration Method (3 iterations) Iterative

📋 GIVEN SYSTEM

```
10x + y + z   = 12
2x + 10y + z  = 13
x + y + 10z   = 12
```

💡 CONCEPT USED

The Jacobi Method is an iterative technique. We rearrange each equation to isolate one variable, then repeatedly substitute old values to get better and better approximations.

Step 1 – Check Diagonal Dominance

For each row, the diagonal element must be larger than the sum of absolute values of all other elements in that row:

- Row 1: $|10|$ vs $|1|+|1| = 2 \rightarrow 10 > 2$ ✓
- Row 2: $|10|$ vs $|2|+|1| = 3 \rightarrow 10 > 3$ ✓
- Row 3: $|10|$ vs $|1|+|1| = 2 \rightarrow 10 > 2$ ✓

Diagonally dominant → Jacobi method will converge.

Step 2 – Rearrange Each Equation to Isolate the Diagonal Variable

```
From Eq1: x = (12 − y − z) / 10
From Eq2: y = (13 − 2x − z) / 10
From Eq3: z = (12 − x − y) / 10
```

We isolate x from the equation that has the largest coefficient for x (the diagonal). This guarantees convergence when the system is diagonally dominant.

Step 3 – Initial Guess (Iteration 0)

Start with $x^{(0)} = 0$, $y^{(0)} = 0$, $z^{(0)} = 0$ (the trivial starting point)

Step 4 – Iteration 1: Substitute Old Values

Using x(0)=0, y(0)=0, z(0)=0:

$x^{(1)} = (12 - 0 - 0) / 10 = 12/10 = \mathbf{1.2}$

$y^{(1)} = (13 - 2(0) - 0) / 10 = 13/10 = \mathbf{1.3}$

$z^{(1)} = (12 - 0 - 0) / 10 = 12/10 = \mathbf{1.2}$

In Jacobi, all three use old values (0,0,0) in this step. All three are computed independently using only the previous iteration's values.

Step 5 – Iteration 2: Substitute Iteration 1 Values

Using x(1)=1.2, y(1)=1.3, z(1)=1.2:

$x^{(2)} = (12 - 1.3 - 1.2) / 10 = 9.5 / 10 = \mathbf{0.95}$

$y^{(2)} = (13 - 2(1.2) - 1.2) / 10 = (13 - 2.4 - 1.2) / 10 = 9.4 / 10 = \mathbf{0.94}$

$z^{(2)} = (12 - 1.2 - 1.3) / 10 = 9.5 / 10 = \mathbf{0.95}$

Step 6 – Iteration 3: Substitute Iteration 2 Values

Using x(2)=0.95, y(2)=0.94, z(2)=0.95:

$x^{(3)} = (12 - 0.94 - 0.95) / 10 = 10.11 / 10 = \mathbf{1.011}$

$y^{(3)} = (13 - 2(0.95) - 0.95) / 10 = (13 - 1.9 - 0.95) / 10 = 10.15/10 = \mathbf{1.015}$

$z^{(3)} = (12 - 0.95 - 0.94) / 10 = 10.11 / 10 = \mathbf{1.011}$

Step 7 – Tabulate All Iterations

Iteration	x	y	z
0 (Initial)	0	0	0
1	1.2000	1.3000	1.2000
2	0.9500	0.9400	0.9500
3	1.0110	1.0150	1.0110
Exact	1.0000	1.0000	1.0000

The values are converging toward the exact solution x = y = z = 1.

✅ APPROXIMATE ANSWER AFTER 3 ITERATIONS

$x \approx 1.011, \quad y \approx 1.015, \quad z \approx 1.011$

📘 **WHAT YOU LEARNED**

Jacobi method oscillates around the true solution, gradually getting closer. The method is guaranteed to converge when the system is diagonally dominant. More iterations give more accurate results.

Example 10 Solve using Gauss-Seidel Method (3 iterations) Iterative

GIVEN SYSTEM (SAME AS EXAMPLE 9)

```
10x + y + z   = 12   →   x = (12 − y − z) / 10
2x + 10y + z  = 13   →   y = (13 − 2x − z) / 10
x + y + 10z   = 12   →   z = (12 − x − y) / 10
```

DIFFERENCE FROM JACOBI

In Gauss-Seidel, as soon as we compute x, we immediately use that new x value to compute y. Then we use the new x AND new y to compute z. This "use immediately" strategy makes it converge faster.

Step 1 – Initial Guess: x(0) = y(0) = z(0) = 0

Step 2 – Iteration 1 (Using New Values Immediately)

Compute x(1) using old y(0)=0, z(0)=0:

$x^{(1)} = (12 - 0 - 0) / 10 = \mathbf{1.2}$

Compute y(1) using NEW x(1)=1.2, old z(0)=0:

$y^{(1)} = (13 - 2(1.2) - 0) / 10 = (13 - 2.4) / 10 = 10.6/10 = \mathbf{1.06}$

Compute z(1) using NEW x(1)=1.2, NEW y(1)=1.06:

$z^{(1)} = (12 - 1.2 - 1.06) / 10 = 9.74 / 10 = \mathbf{0.974}$

Notice how x(1) is immediately used to find y(1), and both x(1) and y(1) are used to find z(1). This is the key improvement over Jacobi!

Step 3 – Iteration 2

$x^{(2)} = (12 - 1.06 - 0.974) / 10 = 9.966/10 = \mathbf{0.9966}$

$y^{(2)} = (13 - 2(0.9966) - 0.974) / 10 = (13 - 1.9932 - 0.974) /$

```
10 = 10.0328/10 = 1.00328
```

$z^{(2)} = (12 - 0.9966 - 1.00328) / 10 = 9.99952/10 = \mathbf{0.999952}$

Iteration Table – Gauss-Seidel vs Jacobi Comparison

Iteration	x (GS)	y (GS)	z (GS)	x (Jacobi)
0	0	0	0	0
1	1.2000	1.0600	0.9740	1.2000
2	0.9966	1.0033	0.9999	0.9500
Exact	1.0000	1.0000	1.0000	1.0000

Gauss-Seidel reached near-exact at iteration 2, while Jacobi was still at 0.95. Gauss-Seidel is clearly faster!

✅ FINAL ANSWER AFTER 2 ITERATIONS

$x \approx 0.9966$, $y \approx 1.0033$, $z \approx 0.9999$ (converging to 1, 1, 1)

📘 **WHAT YOU LEARNED**

Gauss-Seidel converges significantly faster than Jacobi because it uses the most up-to-date values available at each step. For exam problems, Gauss-Seidel typically requires fewer iterations to reach the required precision.

Example 11 Find rank using Normal Form (Canonical Form) Method Medium-Hard

📋 **GIVEN**

```
    [ 1  2  3 ]
A = [ 0  1  1 ]
    [ 1  3  4 ]
```

Step 1 – Apply Row Operations First

R3 → R3 − R1

[0, 1, 1]

```
[ 1  2  3 ]
[ 0  1  1 ]
[ 0  1  1 ]
```

R3 → R3 − R2

[0, 0, 0]

```
[ 1  2  3 ]
[ 0  1  1 ]
[ 0  0  0 ]
```

Step 2 – Now Apply Column Operations to Get Normal Form

C2 → C2 − 2·C1 (eliminate a12 = 2)

```
[ 1  0  3 ]
[ 0  1  1 ]
[ 0  0  0 ]
```

C3 → C3 − 3·C1 (eliminate a13 = 3)

```
[ 1  0  0 ]
[ 0  1  1 ]
```

```
[ 0   0   0 ]
```

C3 → C3 − C2 (eliminate a23 = 1)

```
[ 1   0   0 ]
[ 0   1   0 ]
[ 0   0   0 ]
```

Step 3 – Read the Normal Form

The normal form is:

```
[ I2 | O ]    =   [ 1   0   0 ]
[  O | O ]        [ 0   1   0 ]
                  [ 0   0   0 ]
```

The identity block is 2 × 2 (I_2). So rank = 2.

FINAL ANSWER

rank(A) = 2 (from Normal Form)

Example 12 **Solve a Homogeneous System AX = 0** Medium

📋 **GIVEN SYSTEM**

```
x + 2y − z = 0
2x + 5y + 2z = 0
x + 4y + 7z = 0
```

💡 **CONCEPT USED**

For homogeneous systems, the trivial solution (0,0,0) always exists. We want to find if non-trivial solutions exist by checking if rank < n.

Step 1 – Write Coefficient Matrix and Apply Row Operations

```
A = [ 1   2   −1 ]
    [ 2   5    2 ]
    [ 1   4    7 ]
```

R2 → R2 − 2·R1

```
[0, 1, 4]
```

R3 → R3 − R1

```
[0, 2, 8]
```

R3 → R3 − 2·R2

```
[0, 0, 0]
```

```
A ~ [ 1   2   −1 ]
    [ 0   1    4 ]
    [ 0   0    0 ]
```

Step 2 – Rank and Free Variables

rank(A) = 2, n = 3. Free variables = 3 − 2 = 1.

Let z = t (free parameter). Then:

From Row 2: y + 4t = 0 → y = −4t

From Row 1: $x + 2(-4t) - t = 0 \rightarrow x - 8t - t = 0 \rightarrow x = 9t$

FINAL ANSWER

$x = 9t,\ y = -4t,\ z = t \quad (t \in \mathbb{R},\ t \neq 0$ for non-trivial)

More Solved Examples (13–20)

Example 13 Find inverse of 2×2 matrix using Adjoint Method **Easy**

GIVEN

```
A = [ 3   1 ]
    [ 5   2 ]
```

Step 1 – Find det(A)

det(A) = (3)(2) − (1)(5) = 6 − 5 = **1**. Since det ≠ 0, inverse exists.

Step 2 – Find Adjoint of 2×2 Matrix

For a 2×2 matrix [a b; c d], the adjoint is simply [d −b; −c a].

adj(A) = [2 −1 ; −5 3]

For a 2×2 matrix, the adjoint is obtained by swapping the diagonal elements and changing the sign of the off-diagonal elements. This is the shortcut for 2×2 only.

Step 3 – Apply Formula

A^{-1} = (1/det) × adj(A) = (1/1) × [2 −1; −5 3]

ANSWER

A^{-1} = [2 −1 ; −5 3]

Verification

A × A^{-1} = [3·2+1·(−5) 3·(−1)+1·3; 5·2+2·(−5) 5·(−1)+2·3] = [6−5 −3+3; 10−10 −5+6] = [1 0; 0 1] = I ✓

Example 14 Rank of matrix with parameter 'k' – find k for given rank Medium-Hard

GIVEN

```
A = [ 1   2   3 ]
    [ 2   k   6 ]
    [ 3   6   9 ]
```

Find value of k such that rank(A) = 2.

Step 1 – Apply R2 → R2 − 2·R1 and R3 → R3 − 3·R1

```
A ~ [ 1   2     3    ]
    [ 0   k−4   0    ]
    [ 0   0     0    ]
```

After these operations: R2 becomes [2−2, k−4, 6−6] = [0, k−4, 0] and R3 becomes [0, 6−6, 0] = [0,0,0].

Step 2 – Analyze Rank Based on k

- If $k \neq 4$: Row 2 = [0, k−4, 0] is non-zero. So we have 2 non-zero rows → rank = 2.
- If $k = 4$: Row 2 = [0, 0, 0]. Only 1 non-zero row → rank = 1.

For rank(A) = 2, we need $k \neq 4$.

ANSWER

rank(A) = 2 when $k \neq 4$

Example 15 Verify Cayley-Hamilton Theorem for a 2×2 matrix Medium

GIVEN

```
A = [ 1   2 ]
    [ 3   4 ]
```

Verify Cayley-Hamilton Theorem.

CONCEPT

Cayley-Hamilton Theorem states that every square matrix satisfies its own characteristic equation. First find characteristic polynomial, then substitute A into it and verify result = 0.

Step 1 – Find Characteristic Polynomial

Characteristic equation: det(A − λI) = 0

```
det [ 1-λ    2   ] = (1-λ)(4-λ) - (2)(3) = 0
    [  3    4-λ ]

= 4 - λ - 4λ + λ² - 6 = 0
= λ² - 5λ - 2 = 0
```

Step 2 – Cayley-Hamilton says: $A^2 - 5A - 2I = 0$

Compute A^2:

```
A² = A×A = [1·1+2·3  1·2+2·4] = [7   10]
           [3·1+4·3  3·2+4·4]   [15  22]
```

Compute $A^2 - 5A - 2I$:

```
= [7  10] - 5[1  2] - 2[1  0]
  [15 22]    [3  4]    [0  1]

= [7-5-2    10-10-0] = [0  0] ✓
  [15-15-0  22-20-2]   [0  0]
```

✅ RESULT

$A^2 - 5A - 2I = O$ (Zero Matrix). Cayley-Hamilton Verified!

Example 16

Express a matrix as sum of Symmetric and Skew-Symmetric matrices

Easy-Medium

GIVEN

```
A = [ 3    5   ]
    [ 1   −2   ]
```

KEY FORMULA

Any square matrix A can be written as: A = P + Q, where $P = (A + A^T)/2$ (symmetric) and $Q = (A - A^T)/2$ (skew-symmetric).

Step 1 – Find A^T

```
AT = [ 3    1 ]
     [ 5   −2 ]
```

Step 2 – Find $P = (A + A^T)/2$

```
A + AT = [6   6] / 2  →  P = [3   3]
         [6  −4]             [3  −2]
```

P is symmetric: $P^T = P$ ✓

Step 3 – Find $Q = (A - A^T)/2$

```
A − AT = [0    4] / 2  →  Q = [0    2]
         [−4   0]             [−2   0]
```

Q is skew-symmetric: diagonal = 0, $Q^T = -Q$ ✓

✅ ANSWER: A = P + Q

A = [3 3; 3 −2] + [0 2; −2 0]

Example 17 Find rank using sub-matrix / minor method Medium

GIVEN

```
A = [ 1   2   3 ]
    [ 0   0   0 ]
    [ 2   4   6 ]
```

Step 1 – Check if any 2×2 minor is non-zero

Using rows 1 and 3, columns 1 and 2:

```
Minor = | 1   2 | = 1×4 − 2×2 = 4 − 4 = 0
        | 2   4 |
```

Using rows 1 and 3, columns 1 and 3:

```
Minor = | 1   3 | = 1×6 − 3×2 = 6 − 6 = 0
        | 2   6 |
```

All 2×2 minors formed using rows 1 and 3 are zero (because row 3 = 2 × row 1). Row 2 is all zeros.

Step 2 – Check 1×1 minors (individual elements)

$a_{11} = 1 \neq 0$. So at least one 1×1 minor is non-zero.

ANSWER

rank(A) = 1

Example 18 Solve 4×4 system using Gauss Elimination Tough

GIVEN SYSTEM

```
x1 + x2 + x3 + x4 = 4
x1 + 2x2 + 3x3 + 4x4 = 9
x1 + 3x2 + 6x3 + 10x4 = 18
x1 + 4x2 + 10x3 + 20x4 = 35
```

Step 1 – Write Augmented Matrix [A | B]

```
[ 1   1   1   1   |  4 ]
[ 1   2   3   4   |  9 ]
[ 1   3   6   10  | 18 ]
[ 1   4   10  20  | 35 ]
```

Step 2 – Eliminate Column 1 (R2−R1, R3−R1, R4−R1)

```
[ 1   1   1   1   |  4 ]
[ 0   1   2   3   |  5 ]
[ 0   2   5   9   | 14 ]
[ 0   3   9   19  | 31 ]
```

Step 3 – Eliminate Column 2 (R3−2·R2, R4−3·R2)

```
[ 1   1   1   1   |  4 ]
[ 0   1   2   3   |  5 ]
[ 0   0   1   3   |  4 ]
[ 0   0   3   10  | 16 ]
```

Step 4 – Eliminate Column 3 (R4−3·R3)

```
[ 1   1   1   1   |  4 ]
[ 0   1   2   3   |  5 ]
[ 0   0   1   3   |  4 ]
[ 0   0   0   1   |  4 ]
```

Step 5 – Back Substitution

From Row 4: $x4 = 4$

From Row 3: $x3 + 3(4) = 4 \rightarrow x3 = 4 - 12 = -8$

From Row 2: $x2 + 2(-8) + 3(4) = 5 \rightarrow x2 - 16 + 12 = 5 \rightarrow x2 = 9$

From Row 1: $x1 + 9 + (-8) + 4 = 4 \rightarrow x1 + 5 = 4 \rightarrow x1 = -1$

FINAL ANSWER

$x1 = -1, x2 = 9, x3 = -8, x4 = 4$

Example 19 Using det(AB) = det(A)·det(B) (Cauchy-Binet Application) Easy

GIVEN

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

Verify det(AB) = det(A) × det(B).

Step 1 – Compute det(A) and det(B)

det(A) = 1×4 − 2×3 = 4 − 6 = **−2**

det(B) = 2×3 − 0×1 = 6 − 0 = **6**

det(A) × det(B) = (−2)(6) = **−12**

Step 2 – Compute AB

$$AB = \begin{bmatrix} 1\times2+2\times1 & 1\times0+2\times3 \\ 3\times2+4\times1 & 3\times0+4\times3 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 10 & 12 \end{bmatrix}$$

Step 3 – Compute det(AB)

det(AB) = 4×12 − 6×10 = 48 − 60 = **−12**

RESULT

det(AB) = −12 = det(A) × det(B). ✓ Cauchy-Binet Verified!

Example 20 Find inverse of 3×3 matrix using Adjoint Method (Full Detail) Tough

GIVEN

```
    [ 1  0  1 ]
A = [ 2  1  3 ]
    [ 1  1  1 ]
```

Step 1 – Compute det(A) by cofactor expansion along Row 1

```
det(A) = 1 × |1  3| − 0 + 1 × |2  1|
             |1  1|           |1  1|
= 1(1−3) + 0 + 1(2−1)
= 1(−2) + 0 + 1(1)
= −2 + 1 = −1
```

det(A) = −1 ≠ 0 → Inverse exists.

Step 2 – Find ALL 9 Cofactors

Remember sign pattern: (+)(−)(+) / (−)(+)(−) / (+)(−)(+)

C11 = (+) × |1 3; 1 1| = (1−3) = −2

C12 = (−) × |2 3; 1 1| = −(2−3) = 1

C13 = (+) × |2 1; 1 1| = (2−1) = 1

C21 = (−) × |0 1; 1 1| = −(0−1) = 1

C22 = (+) × |1 1; 1 1| = (1−1) = 0

C23 = (−) × |1 0; 1 1| = −(1−0) = −1

C31 = (+) × |0 1; 1 3| = (0−1) = −1

C32 = (−) × |1 1; 2 3| = −(3−2) = −1

C33 = (+) × |1 0; 2 1| = (1−0) = 1

Step 3 – Cofactor Matrix

```
Cofactor Matrix C = [ -2   1   1 ]
                    [  1   0  -1 ]
```

```
                   [ -1   -1    1 ]
```

Step 4 – Adjoint = Transpose of Cofactor Matrix

```
adj(A) = CT = [ -2    1   -1 ]
              [  1    0   -1 ]
              [  1   -1    1 ]
```

Step 5 – Compute A^{-1} = (1/det) × adj(A)

```
A-1 = (1/-1) × [ -2    1   -1 ] = [  2   -1    1 ]
               [  1    0   -1 ]   [ -1    0    1 ]
               [  1   -1    1 ]   [ -1    1   -1 ]
```

FINAL ANSWER

A^{-1} = [2 −1 1 ; −1 0 1 ; −1 1 −1]

Examples 21–30: Advanced Problems

Example 21 Rank of a matrix with zero row **Easy**

GIVEN

```
A = [2  4  1; 0  0  0; 6  12  3]
```

Analysis

Row 2 is all zeros. Row 3 = 3 × Row 1 (check: 6=3×2, 12=3×4, 3=3×1). So after eliminating:

```
A ~ [2  4  1]   ← Only 1 non-zero row
    [0  0  0]
    [0  0  0]
```

ANSWER

rank(A) = 1

Example 22 Matrix Multiplication (2×3 × 3×2) Easy

GIVEN

```
A = [1  2  3]    B = [1  4]
    [4  5  6]        [2  5]
                     [3  6]
```

Find AB. Order of A: 2×3. Order of B: 3×2. AB will be 2×2.

Step 1 – Compute Each Element of AB

c_{11} = Row 1 of A · Col 1 of B = 1×1 + 2×2 + 3×3 = 1+4+9 = 14

c_{12} = Row 1 of A · Col 2 of B = 1×4 + 2×5 + 3×6 = 4+10+18 = 32

c_{21} = Row 2 of A · Col 1 of B = 4×1 + 5×2 + 6×3 = 4+10+18 = 32

c_{22} = Row 2 of A · Col 2 of B = 4×4 + 5×5 + 6×6 = 16+25+36 = 77

ANSWER

AB = [14 32; 32 77]

Example 23 Check if matrix is Singular or Non-Singular Easy

GIVEN

```
A = [1  2  3; 4  5  6; 7  8  9]
```

Compute det(A)

R2 → R2 − 4R1: [0, −3, −6]. R3 → R3 − 7R1: [0, −6, −12].

After operations: upper triangular form. R3 → R3 − 2·R2: [0, 0, 0].

Since a row of zeros appeared, det(A) = 0 → Matrix is **Singular**.

ANSWER

det(A) = 0 → Singular Matrix. Inverse does NOT exist.

Example 24 Gauss-Seidel: 4 equations system Tough

GIVEN SYSTEM (DIAGONALLY DOMINANT)

```
10x1 + x2 + x3 + x4 = 12
x1 + 10x2 + x3 + x4 = 13
x1 + x2 + 10x3 + x4 = 14
x1 + x2 + x3 + 10x4 = 15
```

Step 1 – Rearrange (each equation, isolate diagonal variable)

```
x1 = (12 − x2 − x3 − x4)/10
x2 = (13 − x1 − x3 − x4)/10
x3 = (14 − x1 − x2 − x4)/10
x4 = (15 − x1 − x2 − x3)/10
```

Step 2 – Iteration 1 (start: all = 0)

```
x1 = 12/10 = 1.2
x2 = (13−1.2)/10 = 1.18
x3 = (14−1.2−1.18)/10 = 1.162
x4 = (15−1.2−1.18−1.162)/10 = 1.1458
```

Step 3 – Observe Convergence

The exact solution is x1=0.7407, x2=0.8519, x3=0.9630, x4=1.0741 (approximately — verify by substitution). The values will converge in 4–5 iterations.

AFTER 1ST ITERATION

x1≈1.2, x2≈1.18, x3≈1.162, x4≈1.146 (converging)

Example 25 Rank of 2×4 matrix Easy

GIVEN

```
A = [1  3  2  4; 2  6  4  8]
```

R2 → R2 − 2·R1

[0, 0, 0, 0]. Row 2 = 2 × Row 1.

ANSWER

rank(A) = 1

Example 26 Solve 2-equation system by Matrix Method ($AX = B$, $X = A^{-1}B$) Easy

GIVEN

```
2x + 3y = 8
x + 2y = 5
```

Write in Matrix Form AX = B

```
A = [2  3; 1  2], X = [x; y], B = [8; 5]
```

Find A^{-1}

det(A) = 4 − 3 = 1. adj(A) = [2 −3; −1 2]. A^{-1} = [2 −3; −1 2].

$X = A^{-1}B$

```
X = [2  -3][8] = [16-15] = [1]
    [-1   2][5]   [-8+10]   [2]
```

ANSWER

x = 1, y = 2

Example 27 Find the value of λ for which system has no unique solution *Medium*

GIVEN

$$\lambda x + y = 1$$
$$x + \lambda y = 1$$

System has no unique solution when det(A) = 0

det $[\lambda\ 1;\ 1\ \lambda] = \lambda^2 - 1 = 0 \to \lambda = \pm 1$

Analyze each case

$\lambda = 1$: Both equations become $x + y = 1 \to$ Infinite solutions.

$\lambda = -1$: Equations become $-x + y = 1$ and $x - y = 1 \to$ Adding: $0 = 2 \to$ Inconsistent.

ANSWER

$\lambda = 1 \to$ Infinite solutions; $\lambda = -1 \to$ No solution

Example 28 Elementary row operations: Convert to Identity Easy

GIVEN

```
A = [2  4; 1  3]
```

Reduce to Identity using EROs.

R1 ↔ R2

```
[1  3; 2  4]
```

R2 → R2 − 2·R1

```
[1  3; 0  -2]
```

R2 → R2 × (−1/2)

```
[1  3; 0  1]
```

R1 → R1 − 3·R2

```
[1  0; 0  1] = I ✓
```

ANSWER

Matrix reduced to Identity in 4 steps.

Example 29 Find rank of a 3×4 matrix Medium

GIVEN

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 3 & 4 & 1 & 2 \\ 2 & 2 & 1 & 1 \end{bmatrix}$$

$R_2 \to R_2 - 3R_1$ and $R_3 \to R_3 - 2R_1$

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & -2 & 1 & -1 \\ 0 & -2 & 1 & -1 \end{bmatrix}$$

$R_3 \to R_3 - R_2$

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & -2 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2 non-zero rows → rank = 2

ANSWER

rank(A) = 2

Example 30 Verify A × adj(A) = det(A) × I Medium

GIVEN

```
A = [1  2; 3  4]
```

Compute det(A), adj(A)

det(A) = 4−6 = −2. adj(A) = [4 −2; −3 1].

Compute A × adj(A)

```
= [1×4+2×(-3)  1×(-2)+2×1] = [4-6   -2+2] = [-2  0]
  [3×4+4×(-3)  3×(-2)+4×1]   [12-12  -6+4]   [ 0  -2]

= -2 × [1  0] = det(A) × I ✓
       [0  1]
```

ANSWER

A × adj(A) = det(A) × I = −2I. Property Verified!

Examples 31–45: More Practice

Example 31 Rank of identity matrix I_3 **Easy**

GIVEN

```
I3 = [1 0 0; 0 1 0; 0 0 1]
```

Analysis

The identity matrix is already in echelon form. All 3 rows are non-zero. No row operations needed.

Every n×n identity matrix has rank = n.

ANSWER

$\text{rank}(I_3) = 3$

Example 32 Find Adjoint of a 3×3 matrix Medium

GIVEN

```
A = [2  1  0; 1  3  1; 0  1  2]
```

Compute All Cofactors

```
C11 = (+)|3 1; 1 2| = 6−1 = 5
C12 = (−)|1 1; 0 2| = −(2−0) = −2
C13 = (+)|1 3; 0 1| = 1−0 = 1
C21 = (−)|1 0; 1 2| = −(2−0) = −2
C22 = (+)|2 0; 0 2| = 4−0 = 4
C23 = (−)|2 1; 0 1| = −(2−0) = −2
C31 = (+)|1 0; 3 1| = 1−0 = 1
C32 = (−)|2 0; 1 1| = −(2−0) = −2
C33 = (+)|2 1; 1 3| = 6−1 = 5
```

Cofactor Matrix then Transpose

```
adj(A) = [ 5  −2   1]T = [ 5  −2   1]
         [−2   4  −2]    [−2   4  −2]
         [ 1  −2   5]    [ 1  −2   5]
```

(In this case adj(A) = cofactor matrix because the matrix is symmetric!)

ANSWER

adj(A) = [5 −2 1; −2 4 −2; 1 −2 5]

Example 33 Solve homogeneous system with trivial solution only **Easy**

GIVEN

```
x + 2y = 0
3x + y = 0
```

Find det(A)

$A = [1\ 2;\ 3\ 1]$. $\det(A) = 1 - 6 = -5 \neq 0$.

Since $\det(A) \neq 0$, $\mathrm{rank}(A) = 2 = n$. Only trivial solution exists.

ANSWER

Only trivial solution: $x = 0$, $y = 0$

Example 34 Properties of Rank: rank(A) = rank(A^T) Medium

GIVEN

```
A = [1  2; 3  6]
```

Verify rank(A) = rank(A^T).

Find rank(A)

R2 → R2 − 3R1: [0, 0]. rank(A) = 1.

Find A^T and its rank

```
AT = [1  3; 2  6]
```

C2 → C2 − 3C1: [0, 0]. rank(A^T) = 1.

ANSWER

rank(A) = rank(A^T) = 1 ✓

Example 35 Gauss Elimination – Infinite Solutions Medium

GIVEN

```
x + y + z = 3
2x + 2y + 2z = 6
```

Augmented Matrix Reduction

```
[1  1  1  |  3]    R2→R2−2R1    [1  1  1  |  3]
[2  2  2  |  6]    →              [0  0  0  |  0]
```

$\text{rank}(A) = \text{rank}([A|B]) = 1 < 3 = n$. Infinite solutions.

Free variables: $y = s$, $z = t$. Then $x = 3 - s - t$.

ANSWER

$x = 3-s-t$, $y = s$, $z = t$ ($s,t \in \mathbb{R}$). Infinite solutions.

Example 36 Rank of product of matrices Medium

PROPERTY

rank(AB) ≤ min(rank(A), rank(B)). Verify with a specific example.

```
A = [1 0; 0 0], B = [1 2; 3 4]
```

Compute AB

```
AB = [1 2; 0 0]
```

rank(A) = 1 (only row 1 non-zero in A).

rank(B) = 2 (det(B) = 4−6 = −2 ≠ 0).

rank(AB): Row 2 is zero → rank(AB) = 1.

rank(AB) = 1 ≤ min(1, 2) = 1 ✓

ANSWER

rank(AB) = 1 ≤ min(rank(A), rank(B)) = 1. Property verified.

Example 37 Jacobi Method – 2 equations

Iterative

GIVEN

```
5x + y = 11
x + 5y = 9
```

Rearrange

```
x = (11 - y)/5
y = (9 - x)/5
```

Iterations

Iteration	x	y
0	0	0
1	2.2	1.8
2	(11−1.8)/5 = 1.84	(9−2.2)/5 = 1.36
Exact	1.75	1.45

True solution: 5x+y=11, x+5y=9 → 24x=46 → x=46/24=1.9167, then y=(9−1.9167)/5=1.4167. (Converging.)

ANSWER

Converges to x ≈ 1.917, y ≈ 1.417

Example 38 Identify the type of matrix Easy

GIVEN

```
A = [0   3  -2]
    [-3  0   5]
    [2  -5   0]
```

Check Properties

1. Diagonal elements: all 0. ✓ (required for skew-symmetric)

2. Is $a_{12} = -a_{21}$? $a_{12}=3$, $a_{21}=-3 \rightarrow 3 = -(-3) = 3$ ✓

3. Is $a_{13} = -a_{31}$? $a_{13}=-2$, $a_{31}=2 \rightarrow -2 = -(2)$ ✓

4. Is $a_{23} = -a_{32}$? $a_{23}=5$, $a_{32}=-5 \rightarrow 5 = -(-5)$ ✓

ANSWER

A is a Skew-Symmetric Matrix. $A^T = -A$.

Example 39 — Echelon form of a 3×3 matrix (full working) — Medium

GIVEN

```
A = [2  4  6; 1  3  4; 3  5  8]
```

R1 ↔ R2 (to get pivot = 1)

```
[1  3  4; 2  4  6; 3  5  8]
```

R2 → R2 − 2R1, R3 → R3 − 3R1

```
[1   3   4]
[0  -2  -2]
[0  -4  -4]
```

R3 → R3 − 2R2

```
[1   3   4]
[0  -2  -2]
[0   0   0]
```

ANSWER

rank(A) = 2 (2 non-zero rows)

Example 40 Show that rank(A) ≤ min(m,n) Easy

PROBLEM

If A is a 3×5 matrix, what is the maximum possible rank?

Reasoning

Rank cannot exceed the number of rows (since we need 3 linearly independent rows in a 3-row matrix — can't have more than 3 independent ones).

Rank also cannot exceed the number of columns (5 columns limit the maximum number of independent columns to 5).

So max rank = min(3, 5) = 3.

ANSWER

Maximum rank of a 3×5 matrix = 3

Example 41 Solve using Gauss-Jordan (Full Reduction to RREF) Tough

GIVEN

```
x + y + 2z = 9
2x + 4y − 3z = 1
3x + 6y − 5z = 0
```

Write Augmented Matrix and Reduce (Step by Step)

```
[ 1   1    2 | 9 ]
[ 2   4   −3 | 1 ]
[ 3   6   −5 | 0 ]
```

R2 → R2 − 2R1: [0, 2, −7 | −17]

R3 → R3 − 3R1: [0, 3, −11 | −27]

```
[ 1   1     2 |    9 ]
[ 0   2    −7 | −17 ]
[ 0   3   −11 | −27 ]
```

R3 → R3 − (3/2)R2: [0, 0, −11+21/2 | −27+51/2] = [0, 0, −1/2 | −3/2]

R3 → R3 × (−2): [0, 0, 1 | 3]

```
[ 1   1    2 |   9 ]
[ 0   2   −7 | −17]
[ 0   0    1 |   3 ]
```

R2 → R2 + 7R3: [0, 2, 0 | 4] → R2 → R2/2: [0, 1, 0 | 2]

R1 → R1 − 2R3: [1, 1, 0 | 3]

R1 → R1 − R2: [1, 0, 0 | 1]

```
[ 1   0   0 | 1 ]  → x = 1
[ 0   1   0 | 2 ]  → y = 2
[ 0   0   1 | 3 ]  → z = 3
```

ANSWER

$x = 1, y = 2, z = 3$

Example 42 Gauss-Seidel convergence check Iterative

GIVEN

```
x + 10y = 11
10x + y = 11
```

Can we apply Gauss-Seidel directly?

Check Diagonal Dominance

Row 1: |1| vs |10| → 1 < 10. NOT diagonally dominant in row 1!

We must rearrange: swap equations.

```
10x + y = 11   →  x = (11 − y)/10
x + 10y = 11   →  y = (11 − x)/10
```

Now: Row 1: |10| > |1| ✓ Row 2: |10| > |1| ✓ Diagonally dominant → converges.

Iteration 1 (start x=0, y=0)

$x^{(1)} = (11-0)/10 = 1.1$, $y^{(1)} = (11-1.1)/10 = 0.99$

Converges to $x = y = 1$.

ANSWER

After rearranging for diagonal dominance: $x = 1, y = 1$

Example 43 Rank of Zero Matrix Easy

GIVEN

```
O = [0  0; 0  0]
```

Analysis

All elements are zero. All rows are zero rows. No non-zero rows exist.

ANSWER

rank(O) = 0 for any zero matrix

Example 44 Find A if adj(A) is given Tough

GIVEN

```
adj(A) = [1  0; 0  1] (Identity)
```

What is A?

Key Property

We know: $A = (1/\det(\mathrm{adj}(A))) \times \mathrm{adj}(\mathrm{adj}(A))$.

For a 2×2 matrix: $\mathrm{adj}(I) = I$ and $\det(I) = 1$.

Also: $\mathrm{adj}(kA) \rightarrow$ and for 2×2: $\mathrm{adj}(\mathrm{adj}(A)) = A$ (when $\det(A) \neq 0$).

If $\mathrm{adj}(A) = I$, then using $A \cdot \mathrm{adj}(A) = \det(A) \cdot I$: $A \cdot I = \det(A) \cdot I \rightarrow A = \det(A) \cdot I$.

This means A is a scalar matrix. For the specific case $\mathrm{adj}(A)=I_2$: $A = I_2$ gives $\mathrm{adj}(I_2) = I_2$ ✓.

ANSWER

A = I (Identity Matrix)

Example 45 TOUGH: Complete analysis of a parametric system Exam-Level

GIVEN SYSTEM

```
x + y + z = 6
x + 2y + 3z = 10
x + 2y + λz = μ
```

Discuss the nature of solutions for all values of λ and μ.

Step 1 – Form Augmented Matrix and Reduce

```
[ 1   1    1 | 6 ]
[ 1   2    3 | 10]
[ 1   2    λ | μ ]
```

R2 → R2 − R1: [0, 1, 2 | 4]

R3 → R3 − R1: [0, 1, λ−1 | μ−6]

R3 → R3 − R2: [0, 0, λ−3 | μ−10]

```
[ 1   1    1    | 6      ]
[ 0   1    2    | 4      ]
[ 0   0   λ−3   | μ−10  ]
```

Step 2 – Case Analysis

Case 1: $\lambda \neq 3$

Row 3 has a non-zero element ($\lambda-3 \neq 0$). rank(A) = rank([A|B]) = 3 = n.

→ **Unique solution** for any value of μ.

Back sub: $z = (\mu-10)/(\lambda-3)$, then y, then x.

Case 2: $\lambda = 3$ and $\mu = 10$

Row 3 becomes [0, 0, 0 | 0]. rank(A) = rank([A|B]) = 2 < 3.

→ **Infinite solutions**. One free variable.

$z = t$ (free), $y = 4 - 2t$, $x = 6 - y - z = 6 - (4-2t) - t = 2 + t$.

Case 3: $\lambda = 3$ and $\mu \neq 10$

Row 3 becomes [0, 0, 0 | μ−10] where $\mu-10 \neq 0$.

rank(A) = 2 but rank([A|B]) = 3. They differ!

→ **Inconsistent**. No solution.

✅ COMPLETE ANSWER

$\lambda \neq 3 \rightarrow$ Unique solution (any μ)

$\lambda = 3, \mu = 10 \rightarrow$ Infinite solutions

$\lambda = 3, \mu \neq 10 \rightarrow$ No solution (inconsistent)

📘 **KEY LEARNING**

Parametric system analysis is a very common exam question. The strategy is always: reduce the augmented matrix, identify rows that depend on the parameter, and check all possible cases for the parameter values.

SECTION 15

Exercise Questions

2-Mark Questions (Short Answer)

Q1 — 2M

Define rank of a matrix. What is the rank of a zero matrix?

Q2 — 2M

What is the maximum rank of a 3×5 matrix?

Q3 — 2M

When does a homogeneous system AX = 0 have non-trivial solutions?

Q4 — 2M

State the Rouché-Capelli theorem for consistency.

Q5 — 2M

Define symmetric and skew-symmetric matrices with examples.

Q6 — 2M

State the Cauchy-Binet formula for square matrices.

Q7 — 2M

What is the condition for a matrix to be non-singular?

Q8 — 2M

Find the rank of A = [1 2 3; 2 4 6].

Q9 — 2M

If rank(A) = 2 and A is 3×4, how many free variables in AX = 0?

Q10 — 2M

What is an elementary row operation? List all three types.

Q11 — 2M

Write the formula for inverse using adjoint method.

Q12 — 2M

Define diagonal dominance with respect to iterative methods.

Q13 2M

Find the order of matrix AB if A is 3×2 and B is 2×4.

Q14 2M

What is the adjoint of a 2×2 matrix [a b; c d]?

Q15 2M

If A is a square matrix, prove that A and A^T have the same rank.

Q16 2M

Explain the key difference between Jacobi and Gauss-Seidel methods.

Q17 2M

What is the rank of the n×n identity matrix I_n?

Q18 2M

If det(A) = 0, what can you conclude about rank(A) for n×n matrix?

Q19 2M

Define normal form of a matrix. What does it look like?

Q20 2M

When is A·adj(A) = det(A)·I valid?

Q21 2M

State the property: rank(AB) ≤ min(rank A, rank B).

Q22 2M

Find rank of A = [1 0 0; 0 1 0; 0 0 0].

Q23 2M

Express A = [1 3; 2 4] as sum of symmetric and skew-symmetric.

Q24 2M

What are the two phases of Gauss Elimination?

Q25 2M

Write the iteration formula for Jacobi method for 2 unknowns.

5-Mark Questions (Full-Length Problems)

Q26 5M

Q27 5M

Find the rank of A using row echelon form: A = [1 2 3; 2 3 4; 3 4 5; 4 5 6].

Find the inverse of A = [1 1 1; 4 3 −1; 3 5 3] using Gauss-Jordan.

Q28 5M

Solve: x+2y+3z=6, 2x+4y+z=7, 3x+2y+9z=14 using Gauss Elimination.

Q29 5M

Test consistency and solve: x+y+z=3, x+2y+3z=4, x+4y+9z=6.

Q30 5M

Solve 27x+6y−z=85, 6x+15y+2z=72, x+y+54z=110 by Gauss-Seidel (3 iter).

Q31 5M

Find values of λ for which rank(A) = 2, where A = [1 λ 3; 2 4 6; λ 2 9].

Q32 5M

Find the inverse using adjoint: A = [2 −1 3; 1 2 −1; −1 3 2].

Q33 5M

Reduce A = [1 2 3 4; 2 1 4 3; 3 4 1 2; 4 3 2 1] to normal form and find rank.

Q34 5M

Solve the homogeneous system: x+2y+3z=0, 2x+3y+4z=0, 3x+4y+5z=0.

Q35 5M

Apply Gauss-Seidel to solve: 5x−y=9, −x+5y−z=4, −y+5z=−6 (3 iter).

Q36 5M

Find rank of A = [2 3 4 5; 3 4 5 6; 4 5 6 7; 9 10 11 12].

Q37 5M

Find A^{-1} by Gauss-Jordan for A = [0 1 2; 1 2 3; 3 1 1].

Q38 5M

Show that the system x−2y+z=0, 2x+y−z=0, 4x+2y−z=0 has non-trivial solutions. Find them.

Q39 5M

For what values of λ and μ is x+y+z=1, 2x+5y+λz=μ, 3x+2y+λz=μ consistent? Find solutions.

Q40 5M

Use Jacobi method to solve: 10x−y=9, −x+10y−2z=20, −2y+10z=−18. Perform 3 iterations.

Q41 5M

Find the rank of A = [1 2 −1 4; 2 4 3 5; −1 −2 6 −7] using normal form.

Q42 5M

Solve: $2x+y+4z=12$, $4x+11y-z=33$, $8x-3y+2z=20$ using Gauss Elimination.

Q43 5M

If A = [1 0 1; 2 1 0; 3 2 1], find A^{-1} and verify $A \cdot A^{-1} = I$.

Q44 5M

Solve by Gauss-Jordan: $3x-y+2z=12$, $x+2y+3z=11$, $2x-2y-z=2$.

Q45 5M

Using Cauchy-Binet, if $\det(A)=3$ and $\det(B)=-2$, find $\det(A^2B^T)$.

Q46 5M

Test if $x+y-z=1$, $4x+2y-2z=3$, $2x+y-z=3$ is consistent. If so, solve it.

Q47 5M

Perform 3 Gauss-Seidel iterations on: $20x+y-2z=17$, $3x+20y-z=-18$, $2x-3y+20z=25$.

Q48 5M

Express A = [3 5 7; 2 4 9; 1 6 8] as sum of symmetric and skew-symmetric matrix.

Q49 5M

Find rank of the matrix obtained after multiplying A = [1 2; 3 4] by B = [2 0; 0 0].

Q50 5M

Solve the system by matrix inverse method: $x+2y=3$, $3x+5y=8$, $2x+y=4$ (check consistency first).

SECTION 16

Theorems, Formula Sheet & Quick Revision

Key Theorems

📜 **THEOREM 1 – ROUCHÉ-CAPELLI THEOREM**

Statement: The system AX = B is consistent if and only if rank(A) = rank([A|B]).

Further:

- rank(A) = rank([A|B]) = n → Unique solution
- rank(A) = rank([A|B]) < n → Infinite solutions (n − r free variables)
- rank(A) ≠ rank([A|B]) → Inconsistent, no solution

📜 **THEOREM 2 – RANK PROPERTIES**

- 0 ≤ rank(A) ≤ min(m, n) for m×n matrix A
- rank(A) = rank(A^T)
- rank(kA) = rank(A) for k ≠ 0
- rank(A + B) ≤ rank(A) + rank(B)
- rank(AB) ≤ min(rank(A), rank(B))
- Elementary row/column operations do NOT change rank
- rank(I_n) = n; rank(O) = 0

📜 **THEOREM 3 – INVERSE PROPERTIES**

- $(A^{-1})^{-1} = A$
- $(AB)^{-1} = B^{-1}A^{-1}$ (order reverses)
- $(A^T)^{-1} = (A^{-1})^T$
- A·adj(A) = adj(A)·A = det(A)·I
- $\det(A^{-1}) = 1/\det(A)$
- Inverse exists ⟺ det(A) ≠ 0 ⟺ A is non-singular

📜 **THEOREM 4 – HOMOGENEOUS SYSTEMS**

- $AX = 0$ always has the trivial solution $X = 0$
- Non-trivial solutions exist ⟺ $\text{rank}(A) < n$
- For $n\times n$: non-trivial solutions exist ⟺ $\det(A) = 0$
- If $\text{rank}(A) = r$, there are $n - r$ linearly independent solutions

📜 **THEOREM 5 – ITERATIVE METHOD CONVERGENCE**

Both Jacobi and Gauss-Seidel methods converge if the system is **strictly diagonally dominant**:

$$|a_{ii}| > \Sigma\ |a_{ij}| \text{ for all } j \neq i, \text{ for each row } i$$

Gauss-Seidel converges faster than Jacobi when both converge. Diagonal dominance is sufficient but not necessary for convergence.

📜 **THEOREM 6 – CAUCHY-BINET**

For square matrices A and B of the same order n:

$$\det(AB) = \det(A) \cdot \det(B)$$

Consequences: $\det(A^n) = [\det(A)]^n$, and if A is orthogonal, $\det(A) = \pm 1$.

Complete Formula Sheet

Rank & Echelon Form

Rank (Echelon Form)	= number of non-zero rows in REF
Rank (Normal Form)	= size r of identity block I_r
Max Rank	$\text{rank}(A) \leq \min(m, n)$

Row operation	`Ri → Ri + k·Rj (doesn't change rank)`

Matrix Inverse

Adjoint Method	`A-1 = (1/det(A)) × adj(A)`
Adjoint	`adj(A) = Transpose of Cofactor Matrix`
Cofactor C_ij	`C_ij = (-1)^(i+j) × M_ij`
2×2 Inverse	`[a b; c d]^(-1) = (1/(ad-bc)) [d -b; -c a]`
Gauss-Jordan	`[A	I] → (row ops) → [I	A^(-1)]`
Key property	`A · adj(A) = det(A) · I`

System of Equations AX = B

Matrix Form	`AX = B; A is m×n, X is n×1, B is m×1`
Consistent (unique)	`rank(A) = rank([A|B]) = n`
Consistent (infinite)	`rank(A) = rank([A|B]) = r < n`
Inconsistent	`rank(A) ≠ rank([A|B])`
Free variables	`n - rank(A) when consistent with infinite sols`
Homogeneous	`AX = 0; non-trivial sols when rank(A) < n`

Iterative Methods

Convergence condition	|a_ii| > sum of |a_ij| for j ≠ i
Jacobi formula (xi)	xi^(k+1) = (bi - sum a_ij·xj^(k)) / a_ii
Gauss-Seidel	Use latest available values immediately
Starting guess	Usually x = 0 (zero vector)
Stopping criterion	|x^(k+1) - x^(k)| < tolerance (e.g., 0.001)

Determinants & Cauchy-Binet

2×2 Determinant	det[a b; c d] = ad - bc
Cauchy-Binet (square)	det(AB) = det(A) · det(B)
Power Rule	det(A^n) = [det(A)]^n
Transpose	det(A^T) = det(A)
Scalar Mult	det(kA) = k^n · det(A) for n×n matrix
Singular	det(A) = 0 ⇔ rank(A) < n

⚡ Quick Revision Notes

🔑 KEY IDEAS

- Rank = non-zero rows in REF
- Rank ≤ min(rows, cols)

❌ COMMON MISTAKES

- Forgetting to transpose cofactor matrix

⚡ EXAM SHORTCUTS

- For 2×2 inverse: swap diagonal, negate off-diagonal, divide by det

- Consistent ↔ rank(A) = rank([A|B])
- Inverse exists ↔ det ≠ 0
- Homogeneous always consistent
- GS faster than Jacobi

- Not applying operations to both sides of [A|I]
- Confusing row and column index in a_{ij}
- In GS, using old values instead of new ones
- Not checking diagonal dominance before iterating
- Stopping at REF when RREF is needed

- If a row is multiple of another → rank drops by 1
- If 0 = constant ≠ 0 appears → inconsistent immediately
- Use Gauss-Jordan over back-substitution for clean solutions
- Always verify: plug answer back into original equations

📝 **SUMMARY OF SOLUTION METHODS**

Problem Type	Best Method	Key Step
Find Rank (small matrix)	Row Echelon Form	Count non-zero rows
Find Rank (any size)	Normal Form	Use both row and column ops
Find Inverse (2×2)	Direct formula	Swap, negate, divide by det
Find Inverse (3×3)	Gauss-Jordan or Adjoint	Use augmented [A|I]
Solve AX=B (small)	Gauss Elimination	Forward elim + back sub
Solve AX=B (large)	Jacobi or Gauss-Seidel	Iterate until convergence
Check Consistency	Rouché-Capelli	Compare ranks
Homogeneous AX=0	Rank method	rank < n for non-trivial

🎯 Last-Minute Revision: What's ALWAYS True

Identity matrix rank	`rank(I_n) = n (always)`
Zero matrix rank	`rank(O) = 0 (always)`
Homogeneous system	`Always consistent (X=0 is solution)`
ERO effect on rank	`No change (rank is invariant)`
Singular matrix	`det = 0, rank < n, inverse does NOT exist`
Gauss-Seidel vs Jacobi	`GS always converges ≥ as fast as Jacobi`
Number of free variables	`= n - rank(A) when consistent`
adj(I) = ?	`adj(I_n) = I_n`
det(I) = ?	`det(I_n) = 1`
Cofactor sign at (i,j)	`= (-1)^(i+j) — positive if i+j even`

🎓 Best of Luck in Your Examinations!

Engineering Mathematics – Unit I: Matrices | B.Tech 1st Year

Remember: Mathematics is not about memorizing formulas — it's about understanding the WHY behind every step.